

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 5+ years shipping TypeScript/JavaScript across the stack — Node.js APIs and data models on the back end, React interfaces on the front end. Led the end-to-end migration of 20+ production applications to Firebase/GCP with zero downtime, building a real-time MongoDB↔Firestore sync pipeline on Google Pub/Sub, and delivered Figma-to-React features at agency pace for demanding US clients. Deep in relational and non-relational data modeling, REST API design, and performance optimization at scale (2M+ record workloads). Code at github.com/zac-09. Now applying the same end-to-end ownership to building premium, high-performance web experiences for Fueled's clients.

CORE COMPETENCIES

TypeScript & Node.js

React

REST API Design

Data Modeling (SQL & NoSQL)

Real-time Systems (Pub/Sub)

Performance Optimization

Microservices Architecture

GCP / Firebase

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Owned the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, from data-model design through deployment, reporting directly to the CTO.
- Built a **real-time two-way sync** between MongoDB and Firestore using Node.js and Google Pub/Sub, keeping a live e-commerce system consistent under parallel production traffic.
- Shipped a customer-facing **3D visualisation feature** (video overlay + ffmpeg) end to end, increasing buyer conversion; delivered Express Shipping, adding **\$40 revenue per order**.
- Cut infrastructure cost by **\$5,000/month** by migrating all services off AWS; moved file storage from S3 to GCS via Python.
- Migrated a **Webflow** marketing site from a Parse backend to Firebase without disrupting the design team's workflow.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration pipeline reading from MongoDB and writing to Firestore after complex processing logic.
- Documented the new Firebase SDKs (Node.js & Python) for fellow engineers and trained the Ops team on dashboards and backup procedures.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer (Full Stack) · US Agency, Remote

- Full-stack delivery for a US-based agency with demanding clients at agency pace, reporting directly to the CTO.
- Delivered pixel-accurate **React** UIs from Figma designs under tight timelines; built Node.js/Web3 backends for DeFi applications.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led **performance optimisation** of a PWA delivering real-time stock prices; built jobs querying **2M+ Firestore records** to keep prices current from Morningstar APIs.
- Built responsive UI pages in React and Tailwind; extended REST APIs with Firebase Cloud Functions and Express.js.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected an event-driven **microservices** backend for FIDA Uganda's case-management platform using Apache Kafka, Docker, and Kubernetes; led a team of 4 developers.
- Built a USSD legal-aid service integrating with a Node.js backend via gRPC; shipped a React Native app with Firebase push notifications.

PROJECTS

Real-time Pub/Sub Sync Pipeline Production

Live two-way MongoDB↔Firestore sync running under parallel production traffic during a zero-downtime migration of 20+ apps.

Node.js · Google Pub/Sub · MongoDB · Firestore

Microservices Backend — FIDA Uganda Architecture

Event-driven backend for a legal case-management platform serving NGOs at scale, with clear service boundaries over gRPC.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC

Marketplace Recommendation Engine

Collaborative + content-based filtering in MongoDB for service-provider matching in a marketplace product.

Node.js · MongoDB

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Languages: TypeScript, JavaScript, Python, Go (intermediate)

Frontend: React, React Native, Tailwind, HTML, CSS

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices

Data: MongoDB, Firestore, SQL, Google Pub/Sub, Apache Kafka

Cloud / Infra: GCP, Firebase, AWS, Docker, Kubernetes, Linux, Git

Spoken: English, Kiswahili